

Features:-

Platform independent:-

Once we write python program it can run on any platform without rewriting once again.

Dynamically typed:-

In python we are not required to declare type of variable, whenever we are assigning value based on value type will be allocated automatically, hence python is considered as dynamically typed language.

Interpreted:-

We are not required to compile python program explicitly, internally python interpreter will take care of that compilation.

It support cross platform:-

It support different OS (Operating System) like windows, linux, mac, ubuntu like chrome, firefox, opera mini, uc browser system like Hp, Dell, apple.

It support Dynamic memory allocation:-

Once we initialize the value, immediately in next line we can reassign the value.

* Keywords :-

- keywords are reserved words (fixed words)
 - Predefined words because they have some special meaning.
1. Python keywords are case sensitive.
 2. In python total 35 keywords are present.
 3. We can not assign any value to the keywords.

How to get all keywords in group.
`help('keywords')` or `help("keywords")`

Here is list of python keywords.

False	class	from	except	and
or	continue	global	finally	return
None	def	if	for	as
pass	del	import	lambda	try
True	elif	in	nonlocal	assert
raise	else	is	not	while
async	with	await	yield	break

How to get all keywords in list format

1. `import keyword`
2. `keyword.kwlist`

`['false', 'None', 'True', ...]`

How to check whether given keyword is valid or not.

a. import keyword

b. keyword.iskeyword("keyword name")

eg: import keyword

keyword.iskeyword("true")

o/p: False

* Variables

A variable is a container that holds information (data)

name = "Baby"

name = variable

Baby = value

- It's a name given to a memory location which holds actual value.

- A variable can holds objects of different types.

- All variables should be in lower case as there are more than one word in variable then user separates with underscore and this called conventions.

Syntax :-

Var - name = value

Snake-case:

- Words are written in lowercase and separated by "(_)" .

eg: `student_name = "Rahul"`

CamelCase :-

- first word starts with lowercase and each next word starts with capital letters.

eg: `studentName = "Rahul"`

Pascal Case :-

- Every word starts with capital letter

eg: `StudentName = "Rahul"`

* type()

`type()` is used to find the data type of a variable or value.

`x = 10`

`print(type(x))`

o/p: `<class int>`
object

O/p always in `<class object>` form.

* id()

It is used to find the memory address (identity) of object.

`a = 10`

`print(id(a))`

Example

- Single variable declaration

`a = 10`

`name = "Prince"`

- Multiple variable declaration (Same line)

~~`a = b`~~

`a, b, c = 10, 20, 30`

- Multiple variable - Same value

`a = b = c = 100`

- Dynamic typing (no need to declare type)

`x = 10`

`# int`

- Swapping Variables

`a = 10`

`b = 20`

`a, b = b, a`

`print(a) # 20`

`print(b) # 10`